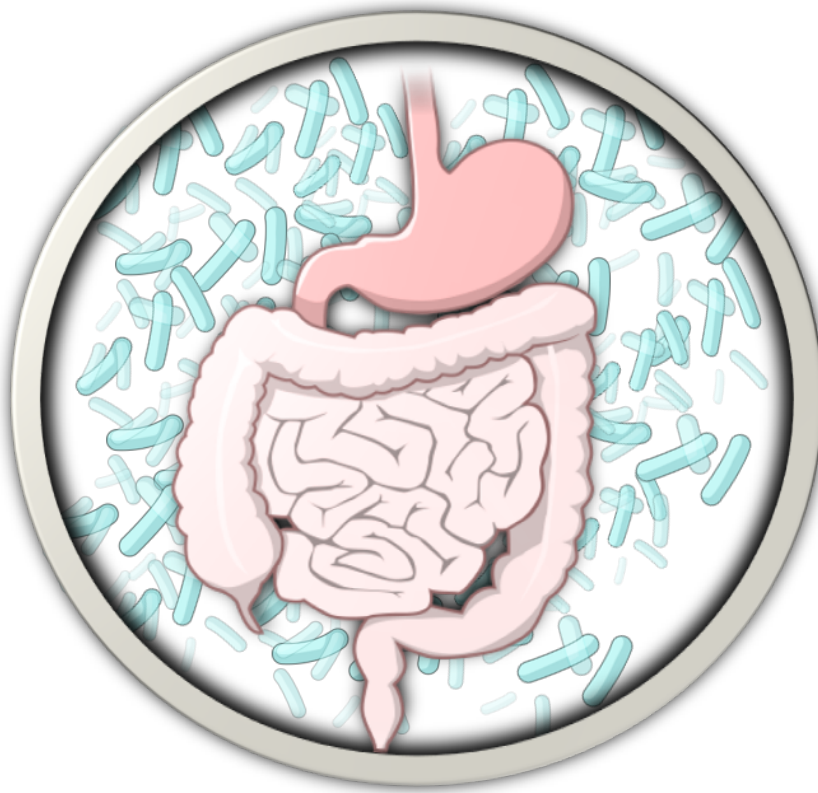




Gut Microbiome Report

Sample ID: Sample001

Disclaimer: These test results are generated using a research-grade testing protocol and are not equivalent to clinical diagnostics performed in accredited hospitals or laboratory settings. They are intended solely for informational purposes and should not be used to guide medical treatment or therapy. If you have any concerns regarding your results, please consult your physician. These reports are not intended for use in any medico-legal context and carry no legal liability.

Index

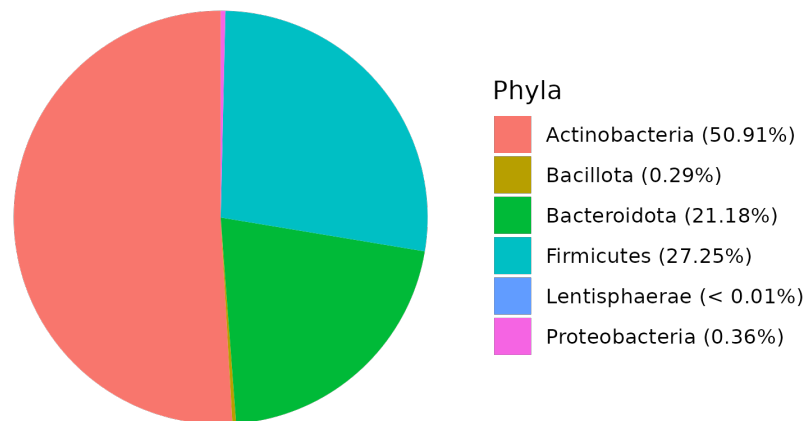
Section	Description
1	Phylum Diversity
2	Firmicutes / Bacteroidetes Ratio
3	Species-level Diversity
3.1	Species Richness
3.2	Evenness Index
3.3	Microbiome Diversity Index
3.4	Distribution of species
4	Probiotic Species and Short Chain Fatty Acid (SCFA) Producers
5	Gut Microbiome Wellness Index: GMWI2

Description

This report presents an analysis of the bacterial diversity and species abundance in your gut microbiome, based on stool sample metagenomic profiling. Our goal is to provide meaningful insights that help you better understand your gut health. The sequencing data was processed using a combination of bioinformatics tools and in-house report generation pipeline to identify and characterize the microbial communities present. We hope this personalized insight empowers you to make more informed decisions about your health and lifestyle.

1) Phylum-Level Diversity of Your Gut Microbiome:

Phylum diversity reflects the range of bacterial phyla found in your gut microbiome. Phyla are broad biological categories that group bacteria based on shared characteristics. In this report, we have assessed both the number and relative abundance of different bacterial phyla present in your gut.

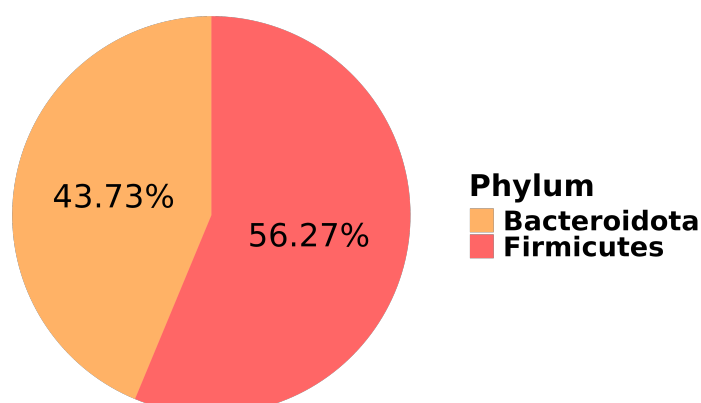


Your gut microbiome profile indicates the presence of 6 distinct phyla. The average range of phyla present in gut microbiome of Indian population is 3-15.*

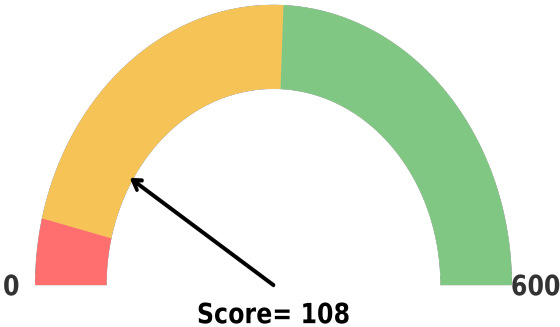
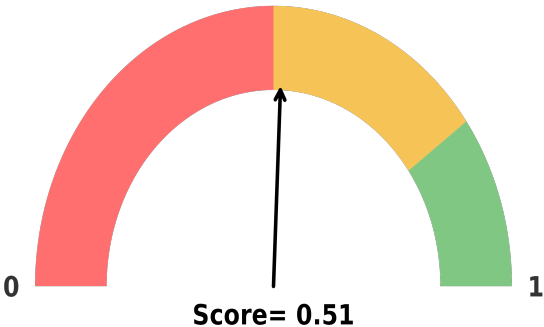
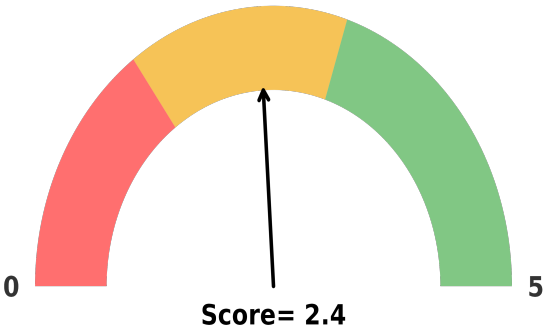
2) Firmicutes and Bacteroidetes Ratio:

Firmicutes (F) and Bacteroidetes (B) are the most common bacterial phyla found in the gut microbiome. The ratio between the abundance of Firmicutes and Bacteroidetes, known as the F/B ratio, is used to measure the relative proportion of these organisms. A balanced proportion (F/B ratio = 1) of these two phyla is vital for maintaining good health, while the majority of the Indian population typically falls within an F/B ratio range of approximately 0.69 to 4.41.*

$$F / B = 1.29$$



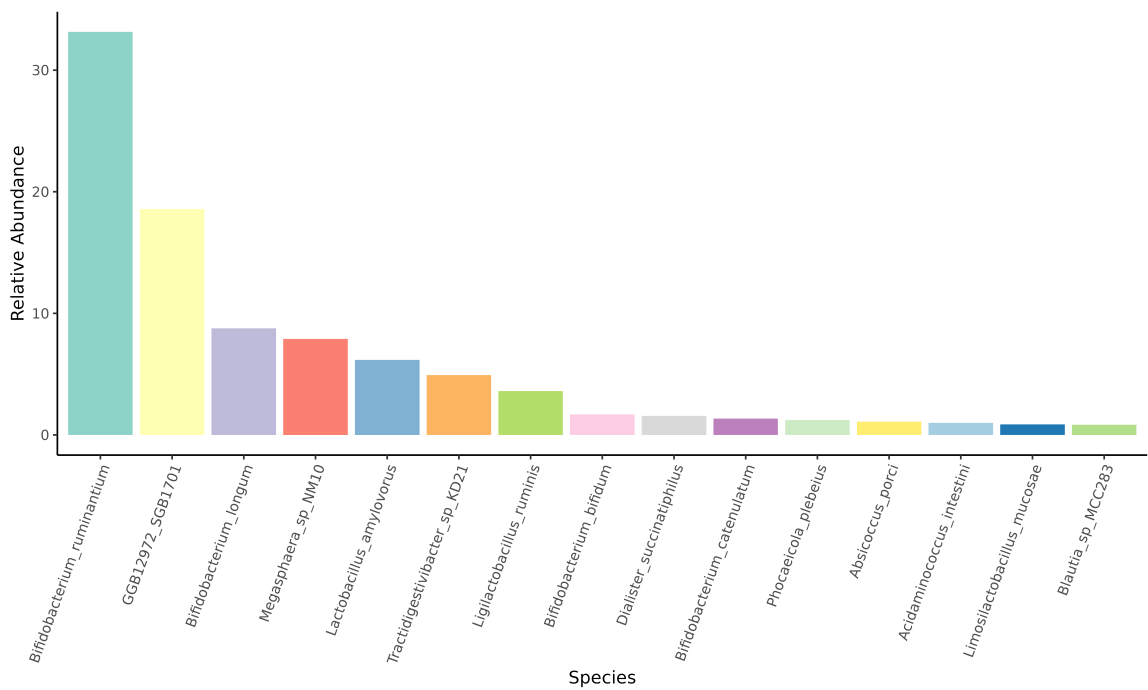
3) Species-level Diversity of Your Gut Microbiome*:

3.1) Microbial Species Richness	Description
 Score= 108	Richness is the total number of microbial species identified within your sample. The number of species found in your gut microbiome is 108 and it is within the average Indian population range.
3.2) Evenness Index	Description
 Score= 0.51	Evenness describes how uniformly all the species are present in your gut microbiome. The value of species evenness in your sample is 0.51 and it is within the average Indian population range.
3.3) Microbiome Diversity Index	Description
 Score= 2.4	The microbiome diversity index is calculated using both richness and evenness. The value of microbiome index of your sample is 2.40 and it is within the average Indian population range.

3.4) Distribution of Species:

The top 15 species in your gut microbiome based on the relative abundance.*

*When a bacterial species has numbers in its name, it signifies a recent discovery that has not been formally named yet.



4) Probiotic Species:

Probiotic species are beneficial microorganisms in the gut that play a vital role in supporting overall health. They aid digestion, strengthen the immune system, protect the intestinal lining, and help reduce inflammation. These microbes also foster a balanced gut environment that supports the growth of diverse microbial communities. Some probiotic species are known to produce Short Chain Fatty Acids (SCFAs)—compounds generated during the fermentation of dietary fiber. SCFAs enhance gut barrier integrity and exhibit anti-inflammatory and immune-regulating effects, contributing to better gut health and a reduced risk of gastrointestinal disorders. (Xiong et al., "Health benefits and side effects of short-chain fatty acids", *Foods*, 2022, Vol. 11, Issue 18, pp. 2863)

This section presents the probiotic species detected in your gut microbiome and SCFA produced by them.

Species	Presence	Short Chain Fatty Acid/s Produced	Description
<i>Alistipes putredinis</i>	+	Acetate, Propionate, Butyrate	It is a bacteria from the phylum Bacteroidetes. It has been shown to have anti-inflammatory properties.
<i>Bifidobacterium adolescentis</i>	-	Acetate, Propionate, Lactate	It is shown to have anti-inflammatory and anti-oxidant properties. It is also used as a probiotic to restore gut health.

<i>Bifidobacterium bifidum</i>	+	Acetate, Lactate	It is a bacteria known for their ability to ferment carbohydrates, help with digestion, nutrient absorption, and fighting off "bad" organisms that can cause disease.
<i>Bifidobacterium breve</i>	+	Acetate, Lactate	It is shown to have antimicrobial activity against human pathogens and immuno-stimulating properties.
<i>Bifidobacterium catenulatum</i>	+	Acetate	It is shown to inhibit harmful pathogens, contribute to the maintenance of a balanced microbiome and support overall gut health and immune function.
<i>Bifidobacterium longum</i>	+	Acetate, Lactate	It is shown to have anti-inflammatory properties, plays role in immunoregulation, and decreasing the infection risk. It is also used as a commercial probiotic to restore gut health.
<i>Blautia wexlerae</i>	+	Not available	It is shown to alter energy metabolism of the host and anti-inflammatory properties. It is also shown to change the composition of the gut microbiome.
<i>Coprococcus catus</i>	+	Butyrate, Acetate	It is a bacteria that plays an important role in fermenting dietary fibers and producing propionate, butyrate and supports gut health.
<i>Faecalibacterium prausnitzii</i>	+	Butyrate, Acetate	It is one of the most abundant bacterial species found in the gut. It is known for it's anti-inflammatory properties and importance in gut health.
<i>Fusicatenibacter saccharivorans</i>	+	Lactate, Acetate, Succinate	It is bacterial species shown to have the ability to reduce inflammation, improve colitis symptoms, and restore gut microbiota balance in patients with ulcerative colitis.
<i>Lactobacillus ruminis</i>	-	Not available	It inhibits harmful pathogens, enhances carbohydrate utilization for energy, and helps maintain a balanced microbial community, thereby supporting digestion and overall immune function.
<i>Prevotella copri</i>	-	Butyrate	It produces short-chain fatty acids (SCFAs) that protect the mucosal barrier and reduce inflammation. It has been associated with reduction of visceral fat and improved glucose metabolism.

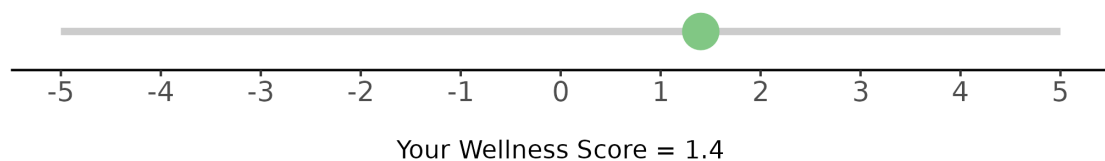
<i>Roseburia faecis</i>	+	Butyrate	It is shown to modulate immune response, reduce inflammation, improve symptoms of irritable bowel syndrome (IBS), and enhance the overall composition of the gut microbiota.
<i>Roseburia hominis</i>	-	Butyrate	It is a bacteria known to produce beneficial short-chain fatty acids (SCFAs) and it has been shown to reduce neuroinflammation.
<i>Roseburia intestinalis</i>	-	Butyrate	It is shown to prevent intestinal inflammation and maintain energy homeostasis. It is also considered as a potential probiotic for treating gut associated disorders.
<i>Ruminococcus bromii</i>	+	Acetate, Propionate, Butyrate	It is bacteria which breaks down complex carbohydrates, producing short-chain fatty acids important for colon health. It also helps promote gut microbiome balance.

Among the Top 16 probiotic species predominantly observed in the Indian population, we identified 11 species in your gut microbiome.*

***Note:** The reference ranges used for interpreting the results are derived from a comprehensive review of publicly available scientific literature on the human gut microbiome.

5) Gut Microbiome Wellness Index 2 (GMWI2):

GMWI2 is a standardized, disease-independent indicator of health status based on gut microbiome taxonomic profiles. A positive GMWI2 score suggests a higher presence of beneficial, health-associated microbes, whereas a negative score indicates a dominance of less beneficial microbial taxa. A GMWI2 value of zero represents a balanced microbial composition, classifying the gut health status as neutral. ([Chang et al., "Gut Microbiome Wellness Index 2 enhances health status prediction from gut microbiome taxonomic profiles", *Nature Communications*, 2024, Vol. 15, Issue 1, pp. 7447](#))



Your gut microbiome wellness score is 1.4 which indicates good gut microbiome.

Interpretation of Your Results:

Your overall gut microbiome analysis indicates a well-balanced and healthy microbiome, supporting optimal gut health.

Contact us

Email: kumardeep.igib@csir.res.in

Address: CSIR-Institute of Genomics & Integrative Biology, South Campus, Mathura Road,
Opp: Sukhdev Vihar Bus Depot, New Delhi 110025